

factor authentication, cloud access security broker, etc, will have to be implemented. Existing security programs would have to be revised to include risk management, business continuity and disaster recovery plans, incident handling, security assessments, security awareness and training, and secure cloud deployment. Implementing life cycle management like logging and monitoring of audit trails, encryption of data, vulnerability assessment, penetration testing and periodic security assessments should be taken into consideration.

Performance considerations of cloud migration strategy

For cloud-hosted applications to meet performance requirements, the bandwidth availability should be investigated. While local networks require a minimum of 100Mbps, the bandwidth availability changes drastically after migration, depending on the tier, package and local Internet bandwidth, making only about 2Mbps to 10Mbps available for the purpose. The organisation will therefore need to understand the data flow for considering appropriate bandwidth requirements for its IT infrastructure.

If not properly done, this could result in latency and unmet business application requirements. Organisations should hence use network modelling tools for evaluating the current application performance, and identify areas of improvement to facilitate better performance after executing cloud migration.

Cost considerations of cloud migration strategy

The cost incurred for cloud migration is yet another important consideration. There are three aspects to this – studying the current on-premise costs, estimation of costs in the cloud, and the costs incurred for the migration activity itself.

An assessment of the direct costs (example: hardware and software costs, software licence costs, maintenance contract costs, warranty, supplies, etc) and indirect costs (example: loss of revenue due to outages) associated with maintaining the on-premise infrastructure over time needs to be carried out. Top cloud providers supply pricing calculators to help with the financial estimation of cloud infrastructure costs.

The cloud migration costs relate to using the Internet or appliances to move data from on-premise to the cloud. In addition, labour costs also have to be accounted for. Having all this information handy during the planning phase helps determine the appropriate strategy for migration.

Cloud migration tools

There are many commercial and open source tools that help with the planning and execution of a cloud migration strategy. They provide services to transfer data to the cloud using the public Internet, private networks and offline transactions. There are also tools that help plan and track the progress of the cloud migration process. AWS, Google and Microsoft have services like AWS Migration Hub, AWS Snowball, Google Cloud Data Transfer Service and Azure Migrate, respectively, to help with migration. Third party vendors like Racemi, RiverMeadow, CloudVelox, etc, also offer products for the same.

The open source community offers tools that are free or low cost alternatives and can be easily customised. Some commercial products are built on top of open source products, and offer limited versions for download. If the migration is not large or complex, open source is a viable option.

Some popular open source tools for cloud migration are Apache Nifi, Clover ETL, Myddleware, Pentaho, Talend Open Studio, etc. To choose the right tool, it is necessary to consider its cost, security features available, supported data models, and the data transformation features to cleanse and enrich the data.

Cloud adoption by enterprises is increasingly common these days. Many firms are leveraging multiple clouds for cost-effective alternatives. Multiple cloud adoption strategies allow for mixing and matching applications to the cloud infrastructure that best suit the specific workload.

The major driver that influences cloud migration today is innovation-driven and customer experience-centric digital transformation. Another major factor is the Internet of Things that aims at connecting devices and humans to collect and act on data in myriad ways. Understanding cloud migration drivers and evaluating various considerations during the planning stages helps ensure that the business adds long-term value by adopting the right cloud migration strategy. **END** 

References

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